



**UNIVERSITY OF SCIENCE
HO CHI MINH CITY**

Security

Adapted from Quinn's Slides

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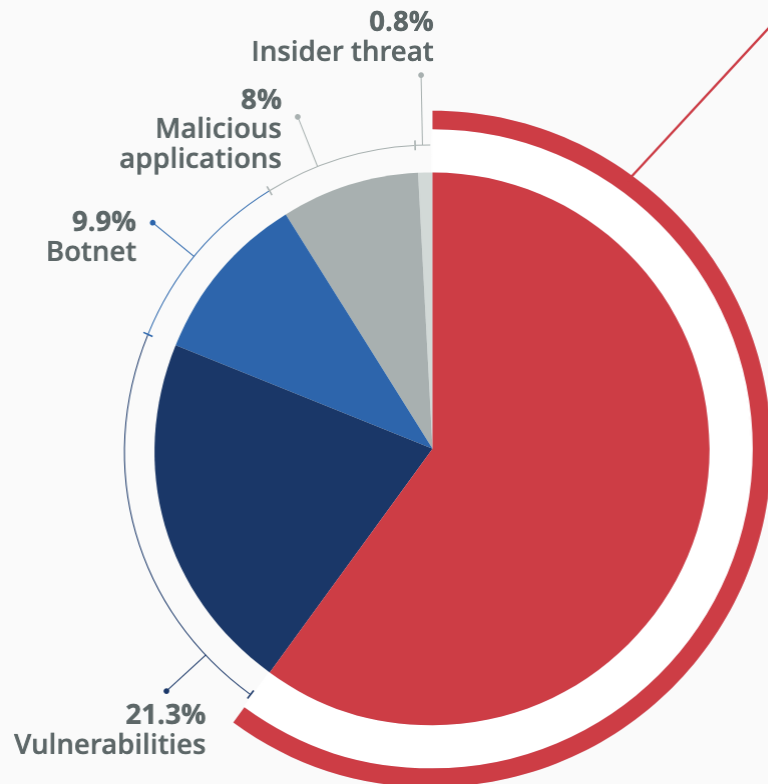
Outline

- Introduction
- Viruses, worms, and Trojan horses
- Phreaks and hackers
- Denial-of-service attacks
- Online voting
- Discussion

Introduction

- Computers getting faster and less expensive
- Utility of computers increasing
 - Email
 - Web surfing
 - Shopping
 - Managing personal information
- Increasing use of computers → growing importance of computer security

Introduction (cont'd)



Most identified initial infection vector

Source: ENISA dataset

60% Phishing

Phishing was the dominant intrusion vector, accounting for approx. 60% of cases, including malspam, vishing, and malvertising. Vulnerability exploitation represented 21.3% of initial access vectors, with 68% leading to malware deployment as a follow-up activity.



Mobile devices and Internet- exposed services and devices, particularly Operational Technology (OT) systems remain high value targets across all types of threats.



State-aligned intrusion sets and cybercriminal operators increasingly leverage AI for productivity and optimisation of their malicious activities.

Source: ENISA
Threat Landscape
2025

Top Cybersecurity Threats in 2025

(Industry consensus from CrowdStrike, ENISA, and Forrester)

Threat

Key Trend in 2025



AI-powered phishing & social engineering

Generative AI enables highly convincing phishing emails, voice cloning, fake identities, and deepfakes (**442% increase in vishing**).



Identity-based attacks

Stolen credentials replace malware as the preferred attack vector (**79% of detections were malware-free**).



Cloud compromise

Increasing abuse of valid cloud accounts and identities for initial access.



Unpatched vulnerabilities

Internet-facing systems remain prime targets for exploitation.



Nation-state cyber espionage

State-sponsored campaigns continue targeting critical infrastructure and strategic industries.



Ransomware & extortion

Data theft and AI-assisted extortion continue to evolve beyond simple encryption attacks.



Supply chain attacks

Compromising software vendors and service providers impacts thousands of downstream organizations.



Deepfakes & synthetic media

AI-generated audio/video increasingly used for fraud, impersonation, and business email compromise.

Viruses

- Virus: piece of self-replicating code embedded within another program (host)
- Viruses associated with program files
 - ❑ Hard disks, CD-ROMS, USB drives
 - ❑ Email attachments
- How viruses spread
 - ❑ CDs, USB drivers
 - ❑ Email
 - ❑ Files downloaded from Internet
- Cure and treatment
 - ❑ Anti-virus software
 - ❑ Continuous updates



Worms

- Worm
 - ❑ Self-contained program
 - ❑ Spreads through a computer network
 - ❑ Exploits security holes in networked computers



The Internet Worm

- Robert Tappan Morris, Jr.
 - Graduate student at Cornell
 - Released worm onto Internet from MIT computer
- Effect of worm
 - Spread to 6,000 Unix computers
 - Infected computers kept crashing or became unresponsive
 - Took a day for fixes to be published
- Impact on Morris
 - Suspended from Cornell
 - 3 years' probation + 400 hours community service
 - \$150,000 in legal fees and fines

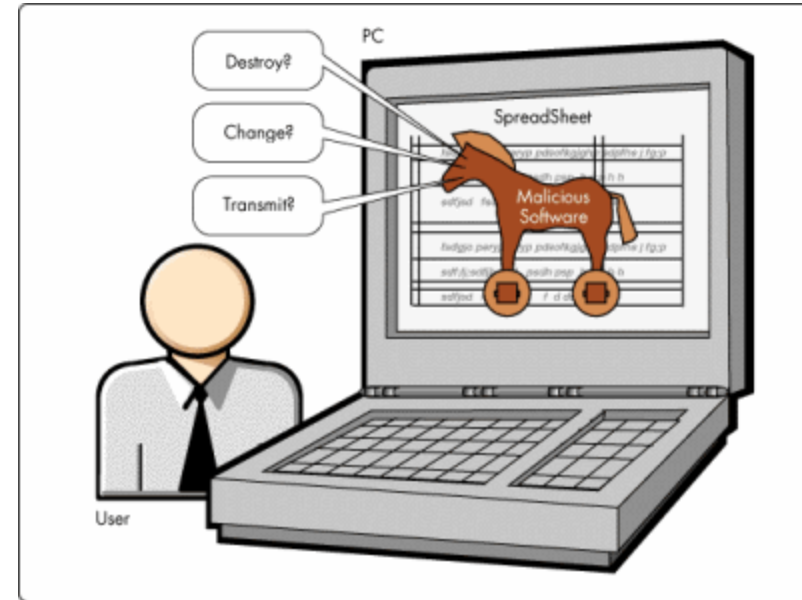
Ethical Evaluation

- Kantian evaluation
 - ❑ Morris used others by gaining access to their computers without permission
- Social contract theory evaluation
 - ❑ Morris violated property rights of organizations
- Utilitarian evaluation
 - ❑ Benefits: Organizations learned of security flaws
 - ❑ Harms: Time spent by those fighting worm, unavailable computers, disrupted network traffic
- Morris was wrong to have released the Internet worm

Trojan Horses

- Trojan horse

- Program with benign capability that masks a sinister purpose



- Remote access Trojan

- Trojan horse that gives attack access to victim's computer
- SubSeven: client and server programs

Bot Networks

- Bot
 - ❑ a software program that responds to commands from a program on another computer
- Some bots support legitimate activities
 - ❑ Internet Relay Chat
 - ❑ Multiplayer Internet games
- Other bots support illegitimate activities
 - ❑ Distributing spam
 - ❑ Collecting person information for ID theft
 - ❑ Distributed denial-of-service attacks



Defensive Measures

- System administrators play key role
- Authorization
 - determining that a user has permission to perform a particular action
- Authentication
 - determining that people are who they claim to be
- Firewall
 - a computer monitoring packets entering and leaving a local area network

Hackers (1/2)

- Original meaning
 - ❑ Explorer
 - ❑ Risk-taker
 - ❑ Technical virtuoso
- Hacker ethic
 - ❑ Hands-on imperative
 - ❑ Free exchange of information
 - ❑ Mistrust of authority
 - ❑ Value skill above all else
 - ❑ Optimistic view of technology



Hackers (2/2)

- Meaning of “hacker” changed
 - ❑ Teenagers accessing corporate or government computers
- Dumpster diving
 - ❑ looking for information in garbage
- Social engineering
 - ❑ Making contact with people in companies
 - ❑ Malicious acts
- Destroying databases
- Stealing confidential personal information
- Attacking websites

Phone Phreaking

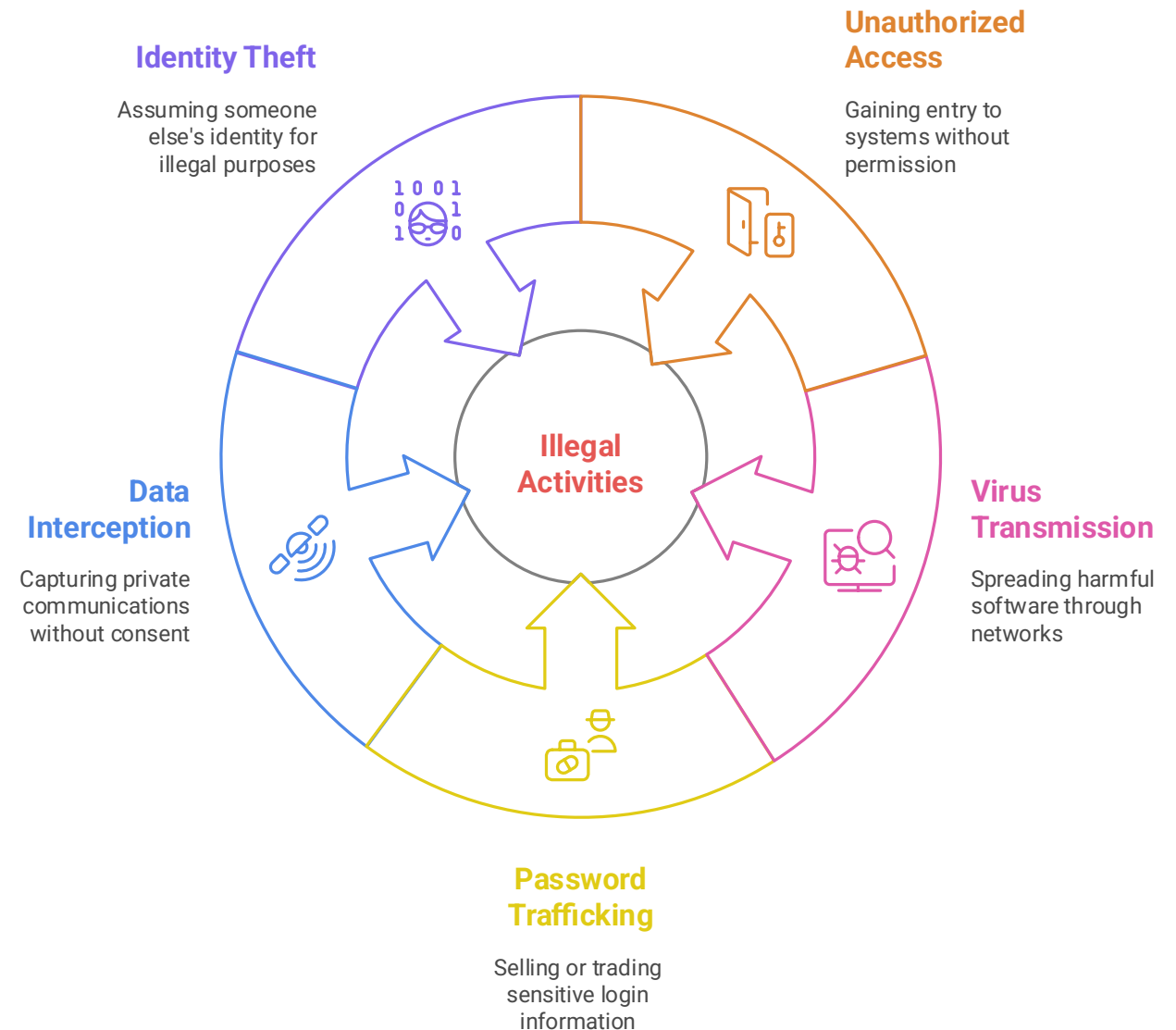
- Phone phreak
 - someone who manipulates phone system to make free calls
- Most popular methods
 - Steal long-distance telephone access codes
 - Guess long-distance telephone access codes
 - Use a “blue box” to get free access to long-distance lines
- Now not popular anymore due to available free phone services



Penalties for Hacking

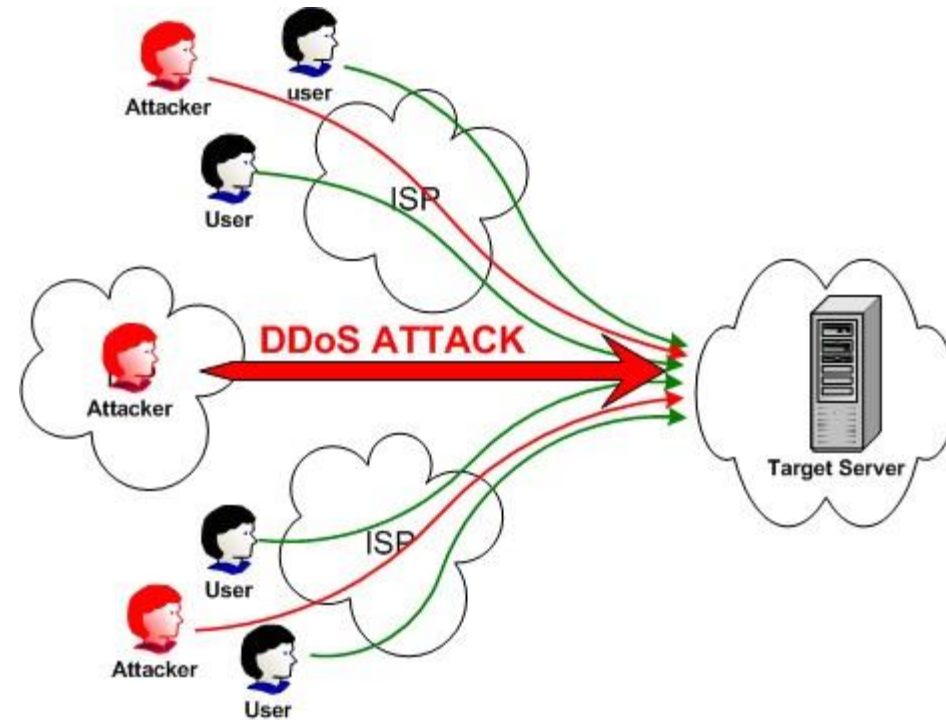
- Maximum penalty:
20 years in prison +
\$250,000 fine (US
law)

Illegal Activities in Digital Space



Denial-of-Service Attacks

- Denial-of-service attack
 - an intentional action designed to prevent legitimate users from making use of a computer service
- Goal of attack
 - disrupt a server's ability to respond to its clients
- About 4,000 Web sites attacked each week



Defensive Measures

- Physical security of server
- Benchmarking
- Disk quota systems
- Disabling unused network services
- Turning off routers' amplifier network capability

Distributed Denial-of-Service Attacks

- Attacker gains access to thousands of computers
- Launches simultaneous attack on target servers
- Defensive measures
 - ❑ Secure computers to prevent hijackings
 - ❑ Check for forged IP addresses
 - ❑ Captcha = Completely Automated Public Turing test to tell Computers and Humans Apart
 - ❑ reCaptcha (Google)

Online Voting

- Voting is cast via Internet
- Benefits
 - ❑ More people would vote
 - ❑ Votes would be counted more quickly
 - ❑ No ambiguity with electronic votes
 - ❑ Cost less money
 - ❑ Eliminate ballot box tampering
 - ❑ Software can prevent accidental over-voting
 - ❑ Software can prevent under-voting



Risks of Online Voting

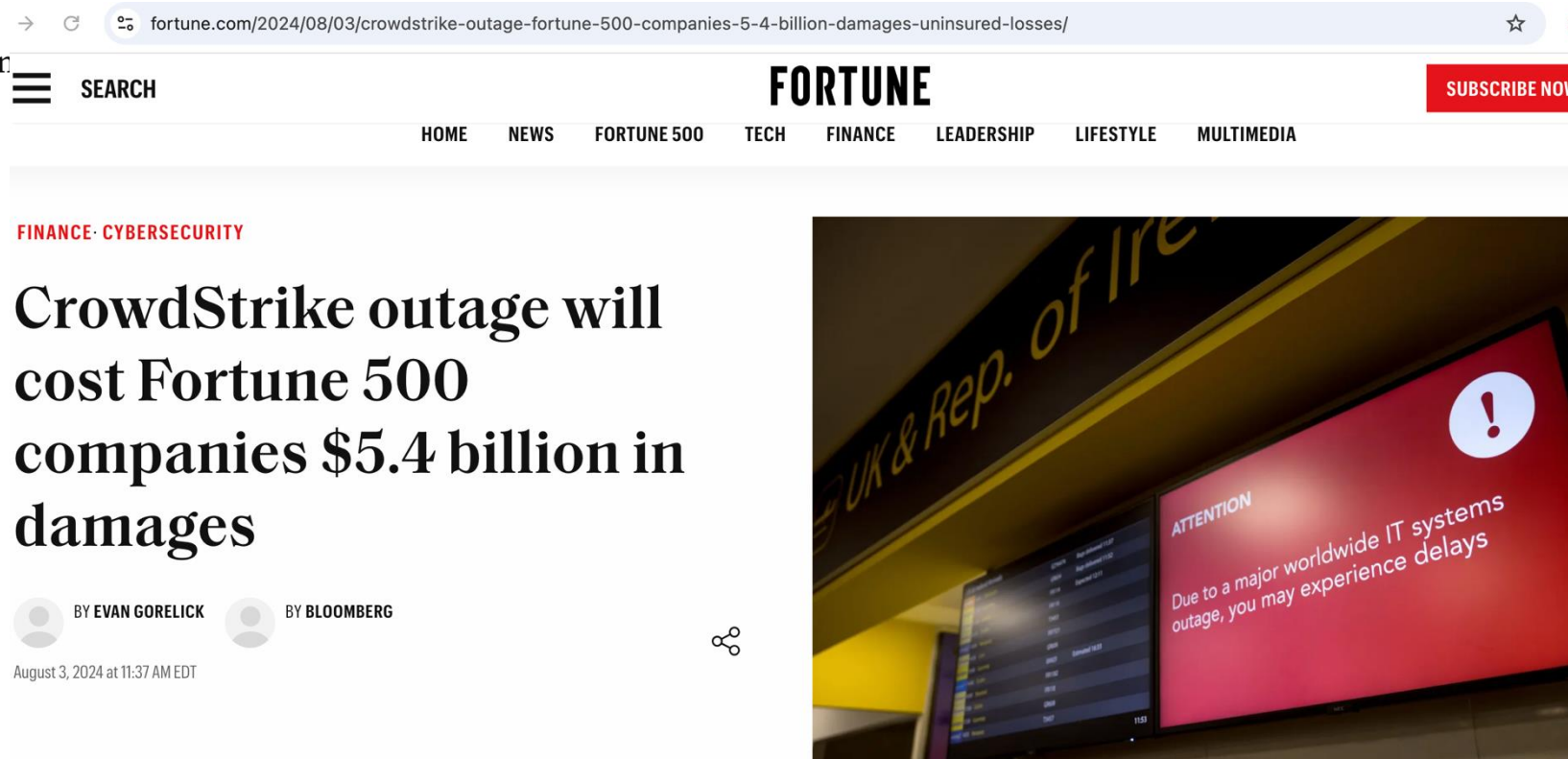
- Gives unfair advantage to those with home computers
- More opportunities for vote selling
- Obvious target for a DDoS attack
- Security of election depends on security of home computers
- Susceptible to phony vote servers
- No paper copies of ballots for auditing or recounts

Kantian Analysis of Online Voting

- The will of each voter should be reflected in that voter's ballot
- The integrity of each ballot is paramount
- Ability to do a recount necessary to guarantee integrity of each ballot
- There should be a paper record of every vote
- Eliminating paper records to save time and/or money is wrong
- Conclusion: strong argument against online voting

CrowdStrike disruption direct losses to reach \$5.4B for Fortune 500, study finds

A report from Parametrix estimates cyber in of losses.



What is the Log4j vulnerability?

What is Log4J?

How hackers use Log4Shell

The impact of Log4Shell

Response to Log4Shell

The persistence of Log4Shell

Mitigation and remediation

A Log4Shell timeline

What is the Log4j vulnerability?

The Log4j vulnerability, also known as [Log4Shell](#), is a critical vulnerability discovered in the Apache Log4j logging library in November 2021. Log4Shell essentially grants hackers total control of devices running unpatched versions of Log4j.

Malicious actors can use the Log4j flaw to run almost any code they want on vulnerable systems.

Researchers consider Log4Shell a “catastrophic” security vulnerability because it is so widespread, Log4J is one of the most widely deployed open source programs in the world,

 cybersecuritydive.com/news/log4j-one-year-later/638416/



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Fear, panic and Log4j: One year later

Fears of catastrophic cyberattacks have thus far failed to materialize. But federal authorities stress threat actors are playing the long game.

Published Dec. 9, 2022

Discussion 1

- In-Class Exercise 31, Chapter 6:
 - ❑ Debate this position: “Those who create nondestructive viruses and worms are doing the computer industry a favor because the patches created to block them make computers more secure. To use an analogy, each virus has the effect of strengthening the immune systems of the computers it targets”
 - ❑ Two groups, for and against this position.
 - ❑ Each group has 5 minutes to prepare arguments
 - ❑ 10 minutes for debate

Discussion 2

- In-Class Exercise 32, Chapter 6:
 - ❑ The University of Calgary offered a senior-level computer science course called “Computer Viruses and Malware”. The course taught students how to write viruses, worms, and Trojan horses. It also discussed the history of computer viruses and taught students how to block attacks. All course assignments were done on closed computer network isolated from the Internet. Some computer security experts criticized the University for offering the course. One researcher said, “No one argues criminology students should commit a murder to understand how a murderer thinks”.
 - ❑ Debate whether the University of Calgary was wrong to offer the course.
 - ❑ Two groups, for and against this position.
 - ❑ Each group has 5 minutes to prepare arguments
 - ❑ 10 minutes for debate